Re-ranking RAG on the GDPR: Failure Modes and Analysis

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Adding a cross-encoder re-ranker put the correct GDPR passage into the LLM's context for 6 of 13 hand-labelled stress-test questions where the bi-encoder alone missed it, and broke 1. It costs about 83 ms per query, under 1% of end-to-end time.

Setup

Both pipelines run over the GDPR (Regulation (EU) 2016/679, 88 pages), split into 305 chunks of 220 words with a 40-word overlap. The bi-encoder is `all-MiniLM-L6-v2` with a FAISS index, the cross-encoder `ms-marco-MiniLM-L-6-v2`, and both pipelines answer with `gemini-3.7-flash`. The test set is 13 questions, each labelled with its gold chunk(s) and grouped by the failure it probes. Timings are averaged over 5 runs per query on a CUDA GPU.

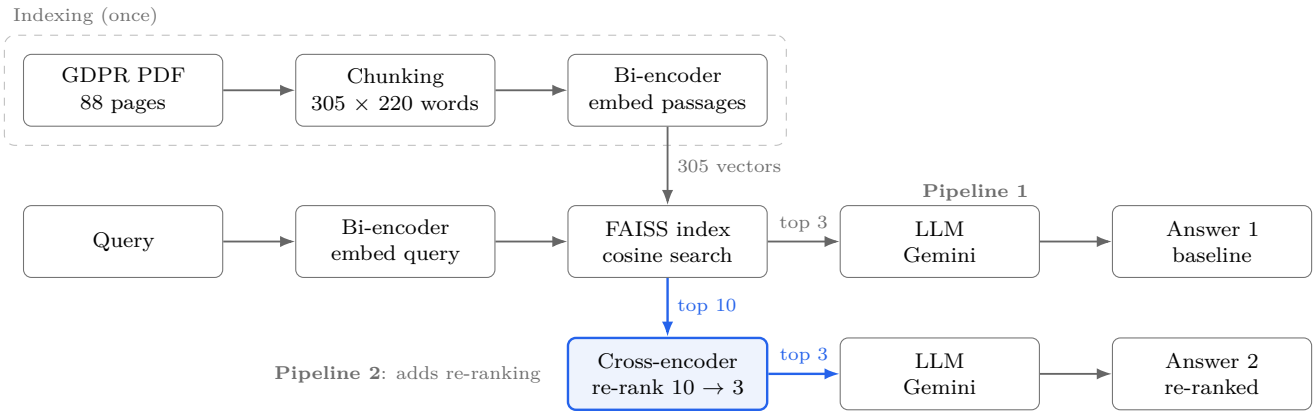


Figure 1: The two pipelines share the index and the first-stage search. Pipeline 2 adds one step: the cross-encoder re-scores the bi-encoder's top 10 and passes its own top 3 to the LLM.

1 Failure mode analysis

Re-ranking restored the gold passage in every required category; one question got worse. Table 1 shows one representative case per failure type. Five more (erasure, dismissal of the data protection officer (DPO), data portability, child consent, a control) had the gold in the top 3 under both pipelines and are left out.

Category	Query	Gold passage (chunk)	Why the bi-encoder stumbles	Outcome
Keyword trap	Does a startup with 10 staff have to maintain records of processing?	Art. 30(5), "fewer than 250 persons" (193)	"Records of processing" matches Recital 82 word for word; the answer says "250 persons", never "startup" or "staff"	Restored
Keyword trap	How long can personal data be stored?	Art. 5(1)(e), storage limitation (146)	Gold never enters the top 10, so the re-ranker has nothing to promote	Both miss
Exception	Must every company appoint a data protection officer?	Art. 37(1), when a DPO is mandatory (205)	Finds Art. 37(4), the general "may designate" rule, and misses the conditions	Restored
Exception	Does a controller always have to notify affected individuals about a data breach?	Art. 34(3), when notice is not required (197, 198)	Gets the general duty (Recital 86) but not the "except when" list	Restored
Negation	Is processing someone's religious beliefs or health information prohibited?	Art. 9(1), the prohibition (153)	"Prohibited" blurs with recitals on derogations; gold sits at rank 10	Restored
Buried	What is the maximum fine for breaching the basic principles of processing?	Art. 83(5), 20 M EUR or 4% (291, 292)	Returns the neighbouring 10 M EUR / 2% tier, Art. 83(4)	Restored
Degraded	Can the Commission ask the Board to examine a matter affecting more than one Member State?	Art. 64(2) (264)	Bi-encoder had it at rank 2; the cross-encoder pushed it to 8	Degraded

Table 1: Representative stress-test questions, one or two per failure type.

Failure hunt and standard retrieval metrics

To check that the hand-picked questions were not cherry-picked, `gemini-3.6-flash` wrote 100 more: an ordinary question and a **wording trap** (borrowing a phrase from the most similar other passage) for each of 50 passages. On ordinary questions re-ranking restored 8 and degraded 2; on wording traps it restored 2 and degraded 5, because the cross-encoder can be pulled toward the passage the phrase came from. 30 of 100 were missed by both pipelines: if the gold is not in the bi-encoder's top 10, re-ranking cannot help.

	13 hand-labelled		100 generated	
Metric	**Baseline**	**Re-ranked**	**Baseline**	**Re-ranked**
Hit@1	0.23	**0.46**	0.34	**0.42**
Hit@3 / Recall@3 (gold reaches the LLM)	0.46	**0.85**	0.60	**0.63**
Hit@10	0.92	0.92	0.77	0.77
MRR@10	0.40	**0.64**	0.50	**0.55**
Precision@3	0.18	**0.31**	0.22	**0.24**
nDCG@10	0.47	**0.64**	0.53	**0.57**

Table 2: Retrieval metrics. Hit@10 cannot change: the re-ranker reorders the same 10 candidates.

Hit@k counts a hit when any gold chunk is in the top k; MRR uses the first gold chunk; Precision@3 and nDCG@10 use every gold chunk (binary relevance). Precision@3 stays low because most questions have only one gold chunk, so 0.33 is its ceiling for them. The hand-labelled set was built to stress the baseline, so the hunt's smaller gain (Hit@3 +3 points) is closer to everyday use. Answer-level metrics (faithfulness, answer relevance) were not scored automatically; Section 3 compares answers by hand.

2 Architecture comparison

A bi-encoder encodes the query and each passage separately and compares two vectors. A cross-encoder reads them as one sequence, so every query token can attend to every passage token.

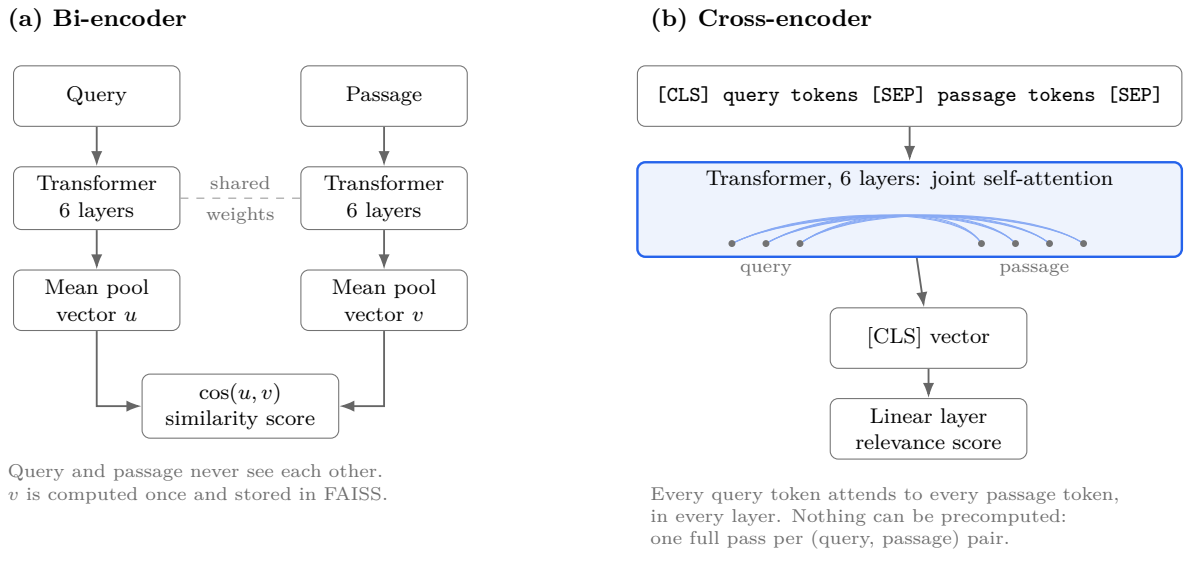


Figure 2: (a) The bi-encoder compresses each text into one 384-d vector on its own, so relevance is a single dot product. (b) The cross-encoder reads the pair as one sequence and outputs a relevance score directly.

The bi-encoder answers "is this passage about the same thing?"; the cross-encoder answers "does this passage answer this question?". Squeezing a passage into one vector means a passage that shares the query's words or topic scores close to the right one, and "not" or "except" barely moves the vector. That is why the cross-encoder pulled Art. 9(1) up from rank 10: it is the sentence that says processing is *prohibited*, while the recitals ranked above it only discuss the topic. The price is speed: the bi-encoder's passage vectors are built once, while the cross-encoder must run one full pass for every candidate at query time.

3 Qualitative and rank shift analysis

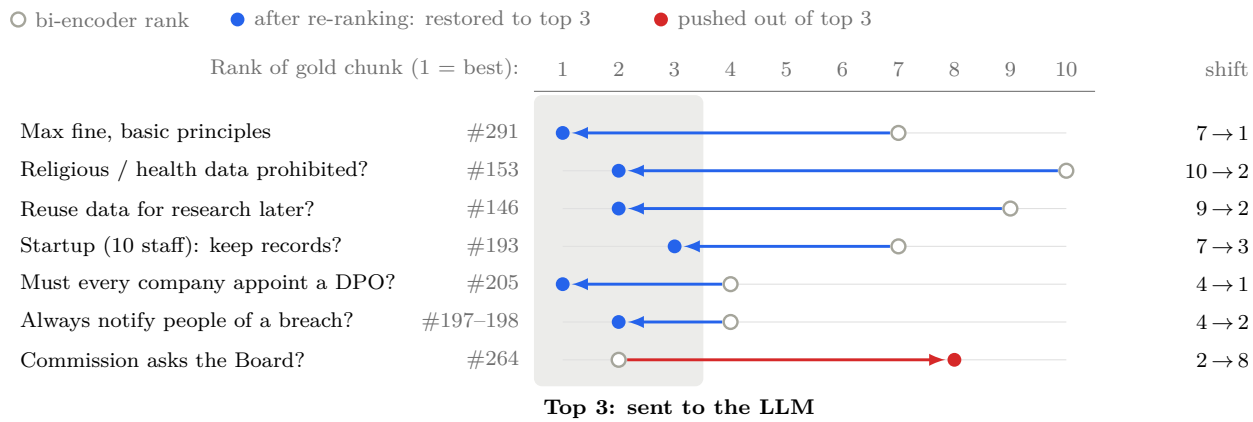


Figure 3: Re-ranking lifted the gold chunk into the top 3 for 6 questions; one fell from rank 2 to 8. The biggest lifts came from deep in the list: Art. 9(1) rose 10 → 2 and the 20 M EUR fine tier 7 → 1.

The baseline never invented a rule outright. Its failures were quieter: it said the context did not cover the question, stated a general rule without its exception, or quoted the wrong tier, each time because the chunk with the deciding clause sat outside the top 3. The research and breach questions failed like the DPO question (rule without its exception), so Table 3 omits them.

Query	Answer 1 (baseline)	Answer 2 (re-ranked)	Baseline failure
Startup with 10 staff: records of processing?	"Not possible to answer"; quotes only Recital 82's general duty	Generally exempt under Art. 30(5) (fewer than 250 persons), unless the processing is risky, not occasional or covers special categories	Omitted the deciding exemption; no answer
Must every company appoint a DPO?	No; cites only Art. 37(4) "may designate"	No; lists the three mandatory cases of Art. 37(1)(a)–(c)	Omitted the conditions that make it mandatory
Religious beliefs or health data prohibited?	Says the context does not list them as prohibited	Yes; Art. 9(1) prohibits it, with the Art. 9(2) exceptions	Misstated the rule: denied a prohibition that exists
Maximum fine for breaching basic principles?	"Not explicitly stated"; gives the 10 M EUR / 2% tier of Art. 83(4)	20 M EUR or 4% of worldwide turnover, Art. 83(5)(a)	Wrong tier on offer; no answer
Commission asks the Board? *(degraded)*	Yes; quotes Art. 64(2)	"Not explicitly stated"; text cuts off	Reverse: re-ranking dropped the gold chunk

Table 3: Answer 1 vs Answer 2 for representative queries.

Because the prompt tells the model to answer only from the context, a dropped chunk produces a cautious "not stated" rather than a confident error, in either pipeline.

4 Latency profiling

Re-ranking makes retrieval 7.5× slower (12.7 → 95.4 ms) but adds only 0.9% to the end-to-end time, because LLM generation takes about 9.2 s either way.

Stage (mean, ms)	Pipeline 1 (baseline)	Pipeline 2 (re-ranked)
Bi-encoder: embed query + FAISS search	12.73 (top 3)	12.73 (top 10)
Cross-encoder: score 10 pairs	—	82.62
Retrieval total	**12.73**	**95.35**
LLM generation (`gemini-3.7-flash`)	9,223.5	9,223.5
End-to-end	**9,236.23**	**9,318.85**

Medians are within 1 ms of the means. Generation time ranged from 5.5 to 18.2 s per answer, driven by answer

length rather than the pipeline. On the CPU (8 threads) the 10-pair re-rank takes about 1 s, still small next to generation.

5 Scaling: why not cross-encode everything?

A cross-encoder over the whole collection costs one transformer pass per passage for every query, and nothing can be computed ahead of time. Scanning this 305-chunk regulation takes 2.65 s; a million passages would take 2.4 hours per question.

Collection size (chunks)	Cross-encoder full scan	Two-stage: re-rank top 10
305 (this GDPR, measured)	2.65 s	82.62 ms
100,000	870 s (14.5 min)	82.62 ms
1,000,000	8,700 s (2.4 h)	82.62 ms
10,000,000	87,001 s (24.2 h)	82.62 ms

- **No precomputation.** The score depends on query and passage together, so nothing can be cached or indexed. Bi-encoder vectors are built once; a FAISS search stays in milliseconds at millions of vectors.
- **Cost grows with the collection.** A full scan is linear in N (8.7 ms per pair); the two-stage design costs a fixed $k = 10$ pairs.
- **Precision only matters at the top.** The bi-encoder only has to get the answer into the top 10 (recall); the cross-encoder spends its expensive attention ordering those 10 (precision).

The catch is that the re-ranker can only promote what the bi-encoder found (30 of 100 hunt questions missed in both). Section 6 widens the first stage with BM25 to target those, at a cost that grows with k, not N.

6 Follow-up: hybrid retrieval and query rewriting

Hybrid. BM25 (over the same chunks, numbers kept as tokens) and the bi-encoder each return their top 20; Reciprocal Rank Fusion ($\sum 1/(60 + \text{rank})$) merges the lists and the fused top 10 go to the same cross-encoder, so re-ranking still costs 10 pairs. Fusion adds 1.3 ms per query. **HyDE.** The local Qwen3.5-4B (Q4_K_M, temperature 0) writes one or two sentences of the passage that would answer the question, in the regulation's wording; the bi-encoder's ranking for that passage is fused as a third list and BM25 searches question and passage together. The cross-encoder still scores the original question.

	13 hand-labelled			100 generated		
After re-ranking	**Dense**	**Hybrid**	**+ HyDE**	**Dense**	**Hybrid**	**+ HyDE**
Hit@1	**0.46**	0.38	**0.46**	0.42	0.43	**0.44**
Hit@3	**0.85**	0.77	0.69	0.63	0.72	**0.73**
Hit@10	0.92	0.92	0.85	0.77	**0.93**	0.92
MRR@10	**0.64**	0.59	0.59	0.55	**0.61**	**0.61**
Added latency	–	1.3 ms	5.6 s	–	1.3 ms	5.6 s

Table 4: Dense is the report's Pipeline 2 and reproduces Table 2. Hybrid and HyDE change only the 10 candidates.

On the hunt, hybrid brought 16 of the 23 questions the bi-encoder missed into the top 10 and lost none; the gold reached the LLM for 12 more questions and 3 fewer. To check that this reaches the answers, the 16 questions whose top 3 changed were answered by Qwen from both top 3s. All 12 gains became correct answers (dense: 10 "not in the context", one wrong, one incomplete). Of the 4 losses, one stayed correct, one became an abstention, one partly right, and one wrong: the startup question, where BM25 matches "records of processing" in Art. 30(1) and pushes the Art. 30(5) exemption ("fewer than 250 persons") from rank 3 to 5, and the model answers that a 10-person startup must keep records. The hand-labelled set was written to defeat keyword overlap, so its small loss

is expected; the hunt is closer to everyday questions. On a second document, the Constitution of India (124 hunt questions), hybrid helps more evenly: re-ranked Hit@3 0.67 to 0.75 and Hit@10 0.77 to 0.91, with 12 questions restored and 2 degraded, and its 12-question case study gains one and loses none.

HyDE adds almost nothing over hybrid on the hunt (Hit@3 +1 point; 6 restored, 5 degraded) and loses a question on the hand-labelled set, for 5.6 s of generation per query. A 4B model's guess at the law is often wrong in the detail that matters: for the startup question it invented the Art. 30 criteria instead of the 250-person threshold, so the rewrite pulls toward the general rule. Hybrid is worth keeping; HyDE with a small model is not.

7 Follow-up: structure-aware chunking

The 220-word windows cut wherever the count falls, often mid-article. The alternative makes each provision (an Article or Recital of the GDPR, a "21. Title.—" article of the Constitution) a unit: small units are packed together and long ones split at their numbered paragraphs, both up to the same 220 words, so only the boundaries change. A third variant embeds and re-ranks each chunk under its heading ("Article 30 Records of processing activities"). Gold chunk ids depend on the chunking, so gold is located in the text instead: each question's snippet is found inside its saved gold chunks, and under any chunking a chunk is gold if it holds at least half of that span. On the windows this returns the saved labels for 106 of 113 GDPR questions; the other 7 add the neighbouring window that shares the snippet through the overlap.

		GDPR (100)		**Constitution (113)**		
Chunking	**Retrieval**	**Hit@3**	**MRR@10**	**Hit@3**	**MRR@10**	**Words to LLM**
Window (current)	dense	0.63	0.55	0.73	0.64	660
Window	hybrid	0.72	0.61	**0.82**	0.70	660
Structure	dense	0.71	0.61	0.65	0.60	490
Structure	hybrid	0.76	0.65	0.81	**0.73**	490
Structure + heading	dense	0.76	0.64	0.68	0.59	495
Structure + heading	hybrid	**0.79**	**0.69**	0.79	0.71	500

Table 5: Failure-hunt questions after re-ranking. Words to LLM: mean length of the top 3 that reach the model. Window + dense reproduces Table 2 (GDPR); its Constitution figures differ from above because 11 questions without a locatable snippet are left out.

On the GDPR, structure-aware chunks help with either retriever: with dense retrieval, Hit@3 rises 8 points (structure) or 13 (structure + heading; 18 questions restored, 5 degraded), and structure + heading + hybrid is the best configuration measured (Hit@3 0.79, MRR 0.69), while sending the model a quarter less text, because structure chunks average 151 words against 220. Headings add 3 to 5 points; the GDPR windows had none. On the Constitution the picture is mixed: with dense retrieval alone structure chunks are worse (Hit@3 0.73 to 0.65, Hit@10 0.81 to 0.70), and with hybrid they tie on Hit@3 but rank the gold first more often (Hit@1 0.56 to 0.62). Its windows were already searched under their article labels, so headings add nothing there. The 13 and 12 hand-labelled questions move by one question either way and are not informative here. Structure-aware chunking is worth it for a cleanly structured regulation, not as a blanket default.

Conclusion

On a legal text where the deciding clause often sits beside a general rule (an exemption, a higher fine tier, a prohibition), a cross-encoder re-ranker is worth its 83 ms: it turned 6 incomplete or wrong answers into correct, cited ones. It is not free of failure: wording traps can mislead it, and it cannot recover passages the bi-encoder never retrieves. Fusing BM25 into the first stage fixes most of those for 1.3 ms (hunt Hit@10 0.77 to 0.93, Hit@3 0.63 to 0.72); rewriting the query with a small local model does not pay for its 5.6 s. Cutting chunks at article boundaries, with headings, lifts the GDPR further (Hit@3 0.79) on a quarter less context, but did not help the Constitution of India, so it suits documents whose structure parses cleanly.

Code

Notebook, saved results, this report and the live app: `https://github.com/Seronic2001/retrieve-ten-keep-three`